# Generating Functions

## Sequences and Series

Let $a_0, a_1, \ldots, a_n, \ldots$ be a sequence. We can represent this using a **power series**:

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots$$

where $a_n$ is the coefficient of $x^n$ in $f(x)$. Note that $x^n$ is a placeholder for $a_n$.

**Basic Idea:** Power series view of sequences gives it algebraic perspective. Functional view could give a succinct representation for series (to be seen in a bit).

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## Recap of Power Series Topics

### 1. Addition:

$$\left(\sum_{n\geq 0} a_n x^n\right) + \left(\sum_{n\geq 0} b_n x^n\right) = \left(\sum_{n\geq 0} (a_n + b_n) x^n\right)$$

### 2. Multiplication:

$$\left(\sum_{n\geq 0} a_n x^n\right) \cdot \left(\sum_{n\geq 0} b_n x^n\right) = \sum_{n\geq 0} \left(\sum_{k=0}^{n} a_k b_{n-k}\right) x^n$$

### 3. Differentiation:

$$\frac{df(x)}{dx} = \frac{d}{dx}\left(\sum_{n\geq 0} a_n x^n\right) = \sum_{n\geq 1} n \cdot a_n x^{n-1}$$

$$= \sum_{n\geq 0} (n+1) a_{n+1} x^n$$

### 4. Integration:

$$\int \sum_{n\geq 0} a_n x^n, dx = \sum_{n\geq 0} \frac{a_n x^{n+1}}{n+1}$$

**Hidden observation:** Linearity of differentiation/integration operators.

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# Familiar Examples

## 1. Geometric Series:

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots = \sum_{n\geq 0} x^n$$

## 2. Modified Geometric:

$$\frac{1}{1-x} \cdot x = x + x^2 + \cdots$$

## 3. More Complex Forms:

$$\frac{1}{(1-x)^2} = \sum_{n=0}^{\infty} (n+1)x^n, \quad \frac{1}{(1-x)^3} = \sum_{n=0}^{\infty} \binom{n+2}{2} x^n$$

$$\frac{1}{(1-x)^k} = \frac{1}{1-x} \times \frac{1}{1-x} \times \cdots \times \frac{1}{1-x}$$

$$= (1 + x + x^2 + \cdots)(1 + x + x^2 + \cdots) \cdots (1 + x + x^2 + \cdots)$$

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# Key Observation

**Observation:** $A_n$ in this case is the number of integral solutions of the equation $y_1 + \cdots + y_k = n$.

**Question:** Show that $A_n = \binom{n+k-1}{k-1}$.

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# Proof Through Sequences and Series

**Rules:**

- $\frac{d}{dx}\left(\frac{1}{f}\right) = -\frac{1}{f^2} \cdot \frac{df}{dx}$
- $\frac{d}{dx}(f(g)) = \frac{d(f(g))}{dg} \cdot \frac{dg}{dx}$

For $\frac{1}{(1-x)^2}$: $\frac{1}{(1-x)^2} = \frac{d}{dx}\left(\frac{1}{1-x}\right)$

On the other hand: $\frac{d}{dx}\left(\sum_{n\geq 0} x^n\right) = \sum_{n\geq 0}(n+1)x^n$

Similarly: $\frac{1}{(1-x)^3} = \frac{d}{dx}\left(\frac{1}{(1-x)^2}\right) = \frac{1}{2}\sum_{n\geq 0}\frac{(n+1)\cdot n\cdot x^{n-1}}{1}$

$$\frac{d}{dx}\left(\frac{1}{(1-x)^{k-1}}\right) = -\frac{d(1-x)^{k-1}}{dx} \cdot \frac{1}{(1-x)^{2k-2}}$$

$$= -k \cdot \frac{(1-x)^{k-2}\cdot(-1)}{(1-x)^{2k-2}} = \frac{k-1}{(1-x)^k}$$

Therefore: $\frac{d}{dx}\left(\frac{1}{(1-x)^{k-1}}\right) \cdot \frac{1}{(1-x)^k} = \frac{1}{(1-x)^k} = \frac{d^{k-1}}{dx^{k-1}}\left(\frac{1}{1-x}\right)$

**Now concentrate on** $\frac{d^{k-1}}{dx^{k-1}}\left(\frac{1}{1-x}\right)$

This is equal to $\frac{d^{k-1}}{dx^{k-1}}\left(\sum_{n\geq 0} x^n\right)$

$$\frac{d^{k-2}}{dx^{k-2}}\left(\sum_{n\geq 1} n\cdot x^{n-1}\right) \to \sum_{n\geq k-1}\frac{n(n-1)\cdots(n-(k-2))\cdot x^{n-(k-1)}}{1}$$

$$= \sum_{n\geq k-1}\frac{n!}{(n-(k-1))!}\cdot x^{n-(k-1)}$$

Setting $m = n-(k-1)$ (Resetting the iterator):

$$\sum_{m\geq 0}\frac{(m+k-1)!}{m!}\cdot x^m$$

**Overall:**

$$\frac{1}{(1-x)^k} = \frac{\frac{d^{k-1}}{dx^{k-1}}\left(\frac{1}{1-x}\right)}{(k-1)!}$$

$$= \sum_{m\geq 0}\frac{(m+k-1)!x^m}{m!(k-1)!} = \sum_{m\geq 0}\binom{m+k-1}{k-1}x^m \quad \square$$

# Generating Function Example

**Question:** What is the generating function of $0, 1 \cdot 2, 2 \cdot 2^2, 3 \cdot 2^3, \ldots$ ?

Start with: $\frac{1}{1-2x} = \sum_{n \geq 0} (2x)^n$

$$\frac{2}{(1-2x)^2} = \frac{d}{dx}\left(\frac{1}{1-2x}\right) = \sum_{n \geq 0} n \cdot 2^n x^{n-1} = \sum_{n \geq 0} (n+1) \cdot 2^{n+1} x^n$$

Writing this we get $1 \cdot 2, 2 \cdot 2^2, 3 \cdot 2^3, \ldots$ but zero does not appear.

**Fix:** Multiply with $x$: $\frac{2x}{(1-2x)^2} = \sum_{n \geq 0} n \cdot 2^n x^n$

$$\frac{2x}{(1-2x)^2}$$

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# Recurrence Relation to Generating Function

$$F_n = F_{n-1} + F_{n-2} \quad \text{where} \quad F(x) = \sum_{n \geq 0} F_n \cdot x^n \,.$$

Multiplying with $x^n$ on both sides:

$$F_n x^n = F_{n-1} x^n + F_{n-2} x^n$$

Sum this from $n \geq 2$:

$$\left(\sum_{n=2}^{\infty} F_n x^n\right) = \left[\sum_{n=2}^{\infty} F_{n-1} x^n\right] + \left[\sum_{n=2}^{\infty} F_{n-2} x^n\right]$$

$$F(x) - F_0 - F_1 x = x \sum_{n=2}^{\infty} F_{n-1} x^{n-1} + x^2 \sum_{n=2}^{\infty} F_{n-2} x^{n-2}$$

$$= x \sum_{n=1}^{\infty} F_n x^n + x^2 \sum_{n=0}^{\infty} F_n x^n$$

$$= x[F(x) - F_0] + x^2 F(x)$$

$$F(x) - F_0 - F_1 x = xF(x) - xF_0 + x^2 F(x)$$

$$F(x)[1 - x - x^2] = F_0(1 - x) + F_1(x)$$

Setting $F_0 = F_1 = 1$:

$$F(x) = \frac{F_0(1-x) + x \cdot F_1}{1 - x - x^2}$$

$$F(x) = \frac{1}{1 - x - x^2}$$

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## Remark

Recall partial fraction decomposition:

$$\frac{1}{1 - x - x^2} = \frac{\phi/\sqrt{5}}{1 - \phi x} - \frac{\hat{\phi}/\sqrt{5}}{1 - \hat{\phi} x}$$

where:

- $\phi = \frac{1+\sqrt{5}}{2}$
- $\hat{\phi} = \frac{1-\sqrt{5}}{2}$

$$F_n = \frac{\phi^{n+1} - \hat{\phi}^{n+1}}{\sqrt{5}}$$

**HW:** If $F_0 = 0$ and $F_1 = 1$ then $F(x) = \frac{x}{1-x-x^2}$

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# More examples

## 1. Number of ways of obtaining change using (infinite) supply of $1, 2, 5$ rupee coins

$$A(x) = \frac{1}{(1-x)(1-x^2)(1-x^5)}$$

## 2. $a_{n+1} = 2 \cdot a_n + 1$

$$\sum_{n \geq 0} a_{n+1} x^n = 2 \sum_{n \geq 0} a_n x^n + \sum_{n \geq 0} x^n$$

$$\frac{A(x)}{x} = 2 \cdot A(x) + \frac{1}{1-x}$$

$$A(x)[1-2x] = \frac{x}{1-x} \Rightarrow A(x) = \frac{x}{(1-2x)(1-x)}$$

$$a_n = 2^{n-1} - 1 \leftarrow A(x) = x\left[\frac{2}{1-2x} - \frac{1}{1-x}\right]$$

**HW:**

1. $a_{n+1} = 2a_n + n$. Find $A(x) \leftarrow \frac{1-2x+x^2}{(1-x)^2(1-2x)}$
2. $a_{n+1} = 2a_n + 3^{n+1} \leftarrow \frac{1}{(1-2x)(1-3x)}$